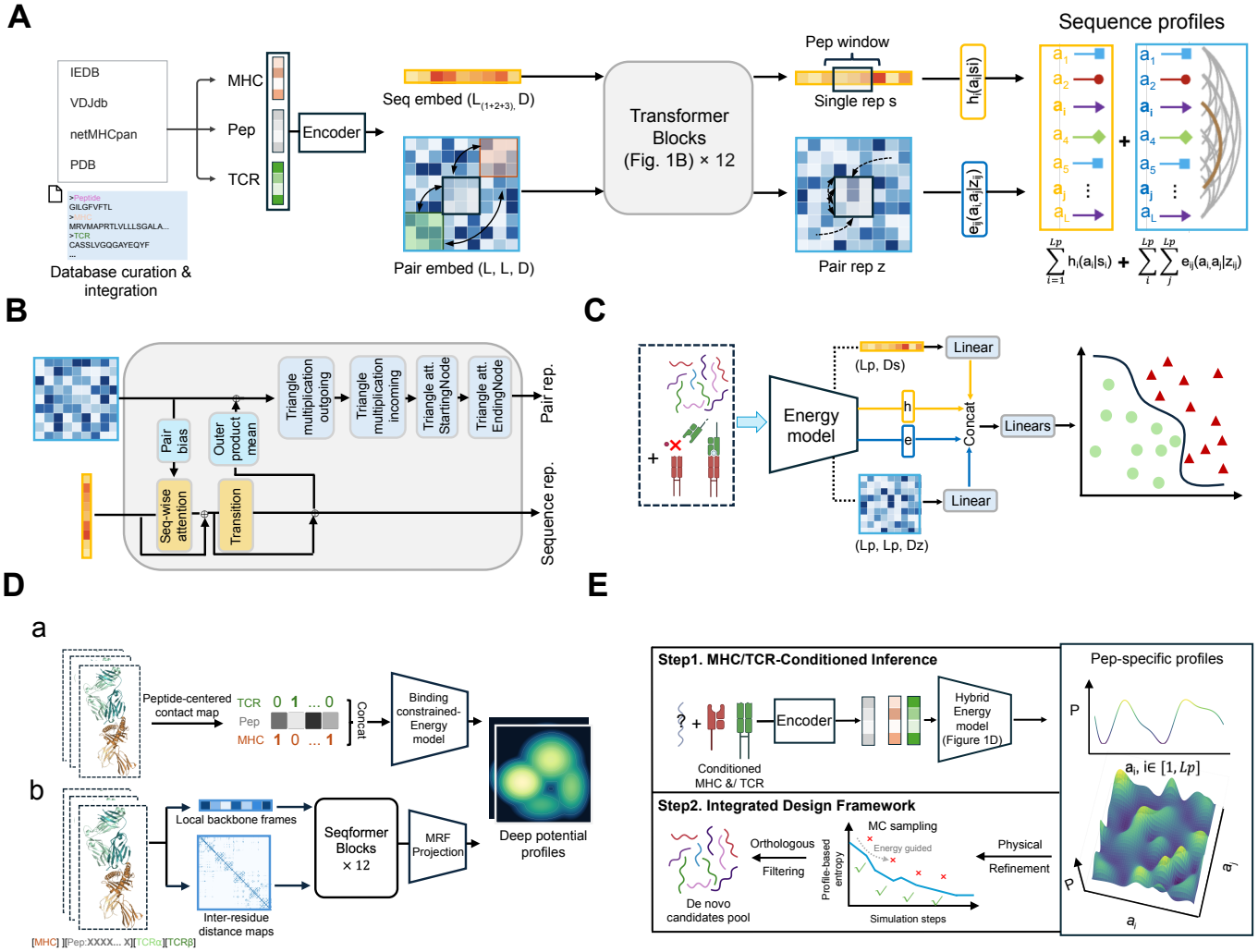

A
Sequence profiles
IEDB
VDJdb
netMHCpan
PDB
MHC
Pep
TCR
Encoder
Database curation & integration
Seq embed (L(1+2+3), D)
Pair embed (L, L, D)
Transformer Blocks (Fig. 1B) × 12
Pep window
Single rep s
Pair rep z
hi(ai|si)
eij(ai,aj|zij)
B
Pair bias
Outer product mean
Triangle multiplication outgoing
Triangle multiplication incoming
Triangle att. StartingNode
Triangle att. EndingNode
Seq-wise attention
Transition
Pair rep.
Sequence rep.
C
Energy model
(Lp, Ds)
(Lp, Lp, Dz)
Linear
Concat
Linears
D
a
Peptide-centered contact map
TCR
Pep
MHC
Concat
Binding constrained-Energy model
Deep potential profiles
b
Local backbone frames
Inter-residue distance maps
Seqformer Blocks × 12
MRF Projection
[MHC][Pep:XXXX... X][TCRα][TCRβ]
E
Step1. MHC/TCR-Conditioned Inference
Encoder
Hybrid Energy model (Figure 1D)
Conditioned MHC &/ TCR
Pep-specific profiles
Step2. Integrated Design Framework
De novo candidates pool
Orthologous Filtering
MC sampling
Energy guided
Profile-based entropy
Simulation steps
Physical Refinement